

1. Introduction

In the summer of 2024, global equity markets experienced one of their sharpest single-week selloffs in years. The trigger was not a recession, a banking crisis, or a geopolitical shock — it was the Bank of Japan raising interest rates by 0.15 percentage points. The outsized reaction revealed something remarkable about the structure of modern financial markets: an enormous volume of capital had been quietly borrowing in Japanese yen at near-zero rates and investing in higher-yielding assets elsewhere, a strategy known as the yen carry trade. When Japanese rates rose and the yen strengthened, the trade unwound simultaneously across thousands of portfolios, transmitting a localised monetary policy decision into a global market shock.

The yen carry trade is sometimes described as a near-frictionless arbitrage — borrowing cheaply in a zero-rate currency and investing in a higher-yielding one, pocketing the differential with apparently limited risk. For much of the period since 2016, when the Bank of Japan introduced Yield Curve Control (YCC) and committed to pinning domestic interest rates at zero indefinitely, this description was not far from the truth. The US-Japan interest rate differential was strictly positive in every single week of our sample, meaning the carry trade was profitable without exception. Leveraged funds were net short yen approximately 75% of the time. The scale was systemic.

But carry trades can stop working — abruptly and violently. When the Federal Reserve began its aggressive hiking cycle in April 2022, the sustainability of the short yen position came into question for the first time. Markets began pricing the inevitable end of YCC well before it formally arrived, and when the BoJ finally abandoned the policy in 2023, the unwind had already been underway for over a year. Understanding precisely how and why this happened — and what it reveals about the true determinants of the yen — is the motivation for this study.

The research question is: what determines the USD/JPY exchange rate, and do these determinants change fundamentally when the BoJ's policy regime shifts? This question matters beyond the yen specifically. It speaks to the broader failure of exchange rate models documented by Meese and Rogoff (1983), and to the question of how central bank yield curve intervention affects the fundamental architecture of currency determination — a question with growing relevance as unconventional monetary policy becomes a standard tool.

To answer it, we use weekly data from January 2016 to December 2025, apply the Bai-Perron structural break test to endogenously identify regime changes, and estimate regime-specific Vector Error Correction Models before and after the identified break. The results show that during the YCC period, the yen's long-run level was anchored entirely by institutional carry trade positioning — macroeconomic fundamentals were irrelevant. After the April 2022 break, the exchange rate transitions to being driven by Japan's yield curve and rate differentials, with the price now actively error correcting toward long-run equilibrium. The carry trade did not just affect the level of the yen — it replaced the entire mechanism by which the yen is priced.

Literature Review

3. Methodology

3.1 Data and Sample Period

This study employs weekly data spanning January 2016 to December 2025 (517 observations), sourced from three providers: interest rate data from FRED; USD/JPY spot rate, VIX, WTI crude oil, DXY, and Nikkei 225 from Yahoo Finance; and weekly Japanese Yen futures positioning data from the CFTC Commitments of Traders (COT) Traders in Financial Futures report. The COT data distinguishes Dealers, Asset Managers, and Leveraged Money, from which net positions, directional intensity, and positioning proportions relative to open interest are constructed. All series are reindexed to weekly frequency and merged via an asof join.

3.2 Feature Engineering

Interest rate differentials are computed at the 3-month, 10-year, and interbank levels as the US rate minus the corresponding Japanese rate. Term spreads (10-year minus 3-month) are constructed for both countries, with the JPY term spread serving as the primary BoJ policy expectations indicator. A Carry-to-Risk ratio is constructed as the 3-month rate differential divided by the VIX, capturing risk-adjusted carry trade attractiveness.

3.3 Stationarity and Transformation

Variables are tested using both the ADF and KPSS tests — a variable is classified as stationary only if ADF rejects the unit root null ($p < 0.05$) and KPSS fails to reject stationarity ($p > 0.05$). Price levels and indices are log-transformed then first-differenced; rates and spreads are first-differenced directly; Japanese interest rates are confirmed I(2) and second-differenced. Endogenous VECM variables are maintained in I(1) levels; exogenous regressors are pre-transformed to I(0).

3.4 Structural Break Detection

The Bai-Perron (1998, 2003) multiple structural break test is applied to log USD/JPY using binary segmentation with a minimum segment length of 52 observations, endogenously identifying break dates without imposing policy event assumptions. Identified breaks are confirmed using Chow tests.

3.5 Cointegration Testing

Pairwise Engle-Granger cointegration tests are conducted between log USD/JPY and all candidate I(1) variables, both for the full sample and within each regime. The Johansen (1988) procedure is then applied within each regime to determine cointegrating rank and estimate the cointegrating vector, with lag order selected by AIC.

3.6 Vector Error Correction Model

The VECM is the primary estimation framework, selected for its ability to simultaneously model long-run cointegrating relationships and short-run dynamics. It is preferred over a

VAR in levels for non-stationary variables and over a VAR in differences as it retains the long-run information that differencing destroys (MacDonald and Taylor, 1994). The specification is:

$$\Delta Y_t = \alpha \beta' Y_{t-1} + \sum \Gamma_i \Delta Y_{t-i} + \Phi X_t + \varepsilon_t$$

where β captures the long-run cointegrating relationship, α captures error correction speeds, and X_t is the vector of stationary exogenous regressors. Separate VECMs are estimated per regime with variable sets determined by within-regime cointegration tests.

3.7 Diagnostic Testing

Residual adequacy is assessed via univariate Ljung-Box tests at lags 1, 4, 8 and 12, the multivariate portmanteau whiteness test for joint autocorrelation, Jarque-Bera normality tests, and Q-Q plots. Non-normality is expected for weekly financial data and treated as motivation for Student's t error distributions rather than evidence of misspecification.

4. Findings

4.1 Data & Descriptive Statistics

4.1.1 Data Sources

This study employs weekly data spanning January 2016 to December 2025, yielding 517 observations. The sample period is chosen deliberately to capture both the full implementation of the Bank of Japan's Yield Curve Control (YCC) policy (introduced September 2016) and its subsequent unwinding from December 2022 onwards, allowing for a meaningful comparison of exchange rate dynamics across the two regimes.

The dependent variable is the **USD/JPY spot exchange rate**, retrieved from Yahoo Finance and expressed in logarithmic form (Log_Price) to ensure stationarity in returns and to allow for percentage change interpretations of coefficients.

Data is drawn from three sources:

Federal Reserve Economic Data (FRED): Interest rate data for both Japan and the United States is sourced from FRED. This includes the 10-year government bond yield, 3-month interbank rate, and overnight interbank rate for both countries. These series form the basis for constructing interest rate differentials and term spreads — central variables in the carry trade and uncovered interest parity frameworks.

Yahoo Finance: In addition to the USD/JPY spot rate, market variables including the CBOE Volatility Index (VIX), the US Dollar Index (DXY), West Texas Intermediate (WTI) crude oil prices, and the Nikkei 225 equity index are sourced from Yahoo Finance. All series are reindexed to the weekly frequency and forward-filled to align with the FRED data.

CFTC Commitments of Traders (COT): Weekly futures positioning data for the Japanese Yen contract traded on the Chicago Mercantile Exchange is sourced directly from the Commodity Futures Trading Commission (CFTC). The Traders in Financial Futures (TFF) report distinguishes between three trader categories: Dealers (financial institutions), Asset Managers (pension funds, mutual funds), and Leveraged Money (hedge funds and CTAs). Net positions, directional intensity, crowding measures, and positioning proportions relative to open interest are constructed for each category.

4.1.2 Variable Construction

Derived Variable	Formula	Economic Interpretation
Rate Differentials		
Rate Differential 3M	US 3M – JPY 3M	Short-end carry trade incentive
Rate Differential 10Y	US 10Y – JPY 10Y	Long-end carry trade incentive
Rate Differential Interbank	US Interbank – JPY Interbank	Overnight funding cost differential
Term Spreads		

US Term Spread	US 10Y – US 3M	Shape of US yield curve
JPY Term Spread	JPY 10Y – JPY 3M	Shape of Japanese yield curve — BoJ policy signal
Term Spread Differential	US Term Spread – JPY Term Spread	Relative yield curve steepness
US 10Y Interbank Spread	US 10Y – US Interbank	US term premium over overnight rate
JPY 10Y Interbank Spread	JPY 10Y – JPY Interbank	JPY term premium over overnight rate
Term 10Y Interbank Differential	US Term Spread – JPY Term Spread	Cross-country term premium differential
Risk-Adjusted Carry		
Carry to Risk	Rate Differential 3M / VIX	Risk-adjusted carry trade attractiveness — high values favour short yen
COT Positioning Metrics		
Dealer Net Proportion	Dealer Net / Open Interest	Dealer directional bias as share of market
Asset Manager Net Proportion	Asset Manager Net / Open Interest	Institutional net positioning as share of market
Levered Net Proportion	Levered Net / Open Interest	Hedge fund net positioning as share of market
Positioning Alignment	Levered Directional Intensity × Asset Manager Intensity	Degree of directional agreement between leveraged and institutional investors

4.1.3 Descriptive Statistics

Variable	Mean	Std Dev	Min	Median	Max	Skewness	Kurtosis
USD/JPY (Price Close)	122.776	17.971	100.251	112.786	161.512	0.672	-1.201
JPY Term Spread	0.211	0.334	-0.289	0.092	0.999	0.711	-0.669
Levered Directional Intensity	-0.221	0.319	-0.736	-0.305	0.655	0.735	-0.428
Asset Manager Net Proportion	-0.015	0.176	-0.288	-0.060	0.491	0.942	0.335
Levered Net (contracts)	-26,461	35,904	-118,526	-28,315	64,179	0.040	-0.402
Asset Manager Net (contracts)	-4,911	45,544	-97,951	-13,755	127,248	0.772	0.152
Rate Differential 3M (%)	2.234	1.769	0.162	1.983	5.472	0.493	-1.104

Rate Differential 10Y (%)	2.371	0.831	0.605	2.450	3.850	-0.378	-0.672
Carry to Risk	0.137	0.115	0.004	0.121	0.429	0.858	-0.055
Oil (USD/barrel)	63.694	17.461	15.060	63.570	122.110	0.274	0.502
VIX	18.496	7.251	9.220	16.730	66.040	2.248	9.103
Nikkei 225	26,872	7,634	15,245	25,860	51,405	0.796	-0.011

Sample period: January 2016 — December 2025, 517 weekly observations.

Several features of the data are worth highlighting. The USD/JPY averaged 122.78 over the sample with negative kurtosis, indicating a relatively flat distribution in levels — masking the dramatic depreciation from ~100 to ~160 that occurred post-divergence. The 3-month rate differential was strictly positive throughout (minimum 0.162), confirming the carry trade was profitable in every week of the sample. Levered Net positions averaged −26,461 contracts with a median of −28,315 and a 75th percentile of approximately zero, meaning leveraged funds were net short yen roughly 75% of the time — direct evidence of the structural carry trade documented by Brunnermeier et al. (2009). The VIX exhibits extreme positive skewness (2.248) and kurtosis (9.103), reflecting episodic risk-off spikes including the COVID-19 shock and the August 2024 carry trade unwind.

4.1.4 Transformations

Original Variable	Transformation	Reason	Output Column Name
LOG + DIFF			
Price Close	log + diff	price level	log Price Close Delta
VIX	log + diff	index level	log VIX Delta
Oil	log + diff	price level	log Oil Delta
Nikkei	log + diff	index level	log Nikkei Delta
DIFF ONLY			
JPY Term Spread	diff	spread	JPY Term Spread Delta
Rate Differential 3M	diff	spread	Rate Differential 3M Delta
Rate Differential 10Y	diff	spread	Rate Differential 10Y Delta
Carry to Risk	diff	ratio	Carry to Risk Delta
Asset Manager Net Proportion	diff	bounded ratio	Asset Manager Net Proportion Delta
Levered Directional Intensity	diff	net ratio	Levered Directional Intensity Delta

4.2. Structural Break Analysis

The Bai-Perron (1998, 2003) multiple structural break test is applied to the log USD/JPY series to endogenously identify structural breaks without imposing a priori assumptions about policy events. The procedure identifies two statistically significant break dates — **April 3, 2022** and **August 6, 2023** — both confirmed by Chow tests at the 1% significance level (Table X).

Break Date	Economic Event	F-Statistic	p-value
April 3, 2022	Federal Reserve begins aggressive hiking cycle	122.76	0.000***
August 6, 2023	Bank of Japan formally abandons YCC	84.95	0.000***

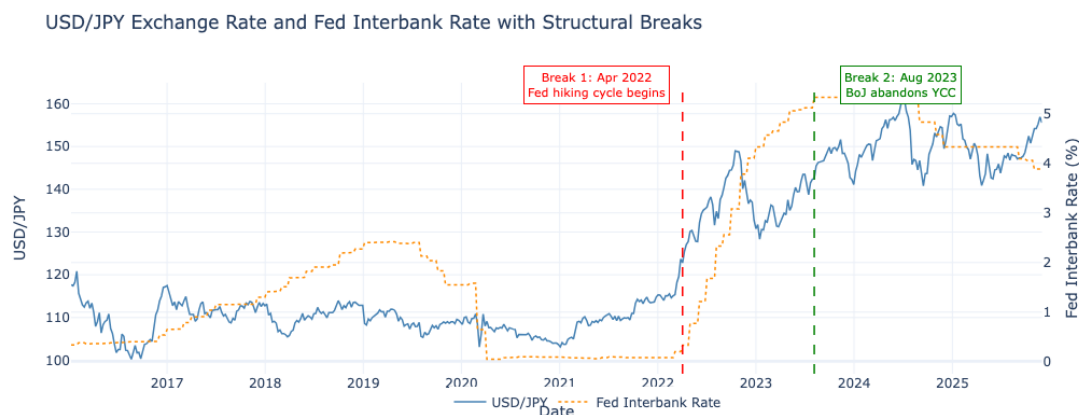


Figure 1: USD/JPY exchange rate and Fed interbank rate (January 2016 — December 2025) with endogenous structural breaks identified by Bai-Perron procedure. Break 1 (April 2022) coincides with the onset of the Federal Reserve hiking cycle; Break 2 (August 2023) coincides with the formal abandonment of BoJ Yield Curve Control.

The first break coincides with the onset of the Federal Reserve’s hiking cycle, which created an divergence between US and Japanese monetary policy. This is theorised to reflect a shift in investor expectations — forward-looking market participants likely began pre-empting either a forced BoJ rate hike or the abandonment of YCC, both of which would trigger an unwind of the structurally short yen carry trade that had been profitable since 2016. The second break corresponds to the BoJ’s formal unwinding of YCC, confirming what markets had anticipated over a year earlier.

While a three-regime framework would provide a richer characterisation, the transitional period between the two breaks contains only 71 observations — insufficient for reliable VECM estimation. This study therefore adopts a **two-regime framework** anchored at April 2022, the more statistically significant break ($F=122.76$):

Regime	Period	Obs	Characterisation
Regime 1	Jan 2016 — Apr 2022	326	YCC stable, carry trade dominates
Regime 2	Apr 2022 — Dec 2025	192	Fed-BoJ divergence, carry trade unwind, fundamentals restored

4.3 Regime 1: YCC Stable (January 2016 — April 2022)

4.3.1 Variable Selection and Cointegration

Pairwise Engle-Granger cointegration tests conducted within Regime 1 identify three variables as sharing a long-run relationship with Log_Price:

- Levered Directional Intensity (p=0.017)
- log Oil (p=0.043),
- Asset Manager Net Proportion (p=0.078).

However, when log Oil is included as an endogenous variable in the VECM, its error correction coefficient is positive and insignificant, indicating it moves away from rather than toward equilibrium — inconsistent with genuine endogeneity. This suggests log Oil is indirectly cointegrated with Log_Price through its relationship with positioning variables rather than directly. Log Oil is therefore retained as an exogenous regressor alongside Carry to Risk, yielding the following specification:

- **Endogenous:** Log_Price, Levered Directional Intensity, Asset Manager Net Proportion
- **Exogenous:** log Oil Delta, Carry to Risk Delta

4.3.2 Lag Selection and Cointegration Rank

The Johansen trace test confirms one cointegrating vector at the 5% significance level (trace statistic=33.34, CV 95%=29.80), with the null of no cointegration ($r \leq 0$) rejected at the 5% level and the null of at most one cointegrating vector ($r \leq 1$) not rejected. The VECM is therefore estimated with $\text{coint_rank}=1$.

Hypothesis	Trace Statistic	CV 90%	CV 95%	CV 99%	Significance
$r \leq 0$	33.341	27.067	29.796	35.463	**
$r \leq 1$	9.362	13.429	15.494	19.935	
$r \leq 2$	2.780	2.706	3.842	6.635	*

* significant at 10%, ** significant at 5%

4.3.3 Cointegrating Vector

The long-run relationship is estimated as

Log_Price = 4.684 – 0.082×Levered Directional Intensity – 0.059×Asset Manager Net Proportion

with both positioning variables significant at the 5% level or better. The negative coefficients confirm that increased net short yen positioning by both leveraged funds and asset managers is associated with yen depreciation, consistent with the carry trade mechanism.

Variable	Coefficient	Std Error	z-stat	p-value	95% CI
Log_Price (β_1)	1.000	—	—	—	—
Levered Directional Intensity (β_2)	0.082	0.013	6.113	0.000***	[0.056, 0.109]
Asset Manager Net Proportion (β_3)	0.059	0.025	2.336	0.020**	[0.009, 0.108]
Constant	-4.684	0.005	-1034.702	0.000***	[-4.693, -4.675]

** significant at 5%, *** significant at 1%

4.3.4 Error Correction Dynamics

Equation	α (ec1)	Std Error	z-stat	p-value	95% CI
Log_Price	-0.097	0.029	-3.301	0.001***	[-0.154, -0.039]
Levered Directional Intensity	-0.646	0.238	-2.719	0.007***	[-1.112, -0.180]
Asset Manager Net Proportion	0.038	0.076	0.505	0.613	[-0.110, 0.186]

*** significant at 1%

Both Log_Price and Levered Directional Intensity error correct significantly, adjusting approximately 9.7% and 64.6% of any deviation per week respectively. Asset Manager Net Proportion does not error correct ($p=0.613$), entering the long-run relationship as a passive anchor rather than an active adjuster. Notably, with correctly specified stationary exogenous variables, Log_Price now error corrects significantly — a finding that was obscured in the earlier non-stationary specification.

4.3.5 Short-Run Dynamics

Panel A: Log_Price equation

Variable	Coefficient	Std Error	z-stat	p-value
Carry to Risk Delta (exog1)	0.175	0.049	3.597	0.000***
log Oil Delta (exog2)	-0.005	0.010	-0.508	0.611
L2.Levered Directional Intensity	0.015	0.006	2.290	0.022**
All other lags	—	—	—	n.s.

Panel B: Levered Directional Intensity equation

Variable	Coefficient	Std Error	z-stat	p-value
L1.Log_Price	-1.992	0.506	-3.938	0.000***
L2.Log_Price	-1.344	0.509	-2.643	0.008***
L1.Asset Manager Net Proportion	0.460	0.194	2.373	0.018**
All other terms	—	—	—	n.s.

Panel C: Asset Manager Net Proportion equation

Variable	Coefficient	Std Error	z-stat	p-value
Carry to Risk Delta (exog1)	-0.350	0.125	-2.797	0.005***
L1.Asset Manager Net Proportion	0.135	0.062	2.196	0.028**
All other terms	—	—	—	n.s.

** significant at 5%, *** significant at 1%

Three short-run findings stand out. First, Carry to Risk Delta is highly significant in the Log_Price equation ($p=0.000$) — short-run changes in risk-adjusted carry directly drive the exchange rate. Second, lagged Log_Price drives leveraged positioning at both one and two week lags ($p=0.000$; $p=0.008$), consistent with trend-following behaviour by hedge funds. Third, Carry to Risk Delta negatively drives Asset Manager Net Proportion ($p=0.005$), meaning improved carry attractiveness reduces asset manager net long yen positioning — these flows were absorbed by BoJ intervention rather than transmitting to the price.

4.3.6 Diagnostics

All Ljung-Box tests pass comfortably across all variables and lags, and the multivariate portmanteau test confirms no joint autocorrelation ($p=0.602$), indicating correct model specification. Jarque-Bera rejects normality for all variables — expected for weekly financial data and confined to tail behaviour as confirmed by the Q-Q plots, which show residuals tracking the normal line closely through the central quantiles with deviations only at the extremes.

Panel A: Univariate Residual Diagnostics

Variable	LB lag 1	LB lag 4	LB lag 8	LB lag 12	JB Stat	JB p-value	Skewness	Kurtosis
Log_Price	0.660	0.132	0.346	0.627	63.25	0.000***	-0.320	2.127
Levered Directional Intensity	0.858	0.933	0.514	0.729	11.24	0.004***	0.063	0.940
Asset Manager Net Proportion	0.915	0.576	0.878	0.964	78.80	0.000***	0.302	2.403

Panel B: Multivariate Portmanteau Test

Test	Test Statistic	Critical Value (5%)	p-value	Result
Whiteness test (lag 12)	94.771	123.225	0.602	Fail to reject H_0

H_0 : residual autocorrelation up to lag 12 is zero. LB = Ljung-Box p-values. JB = Jarque-Bera. *** significant at 1%

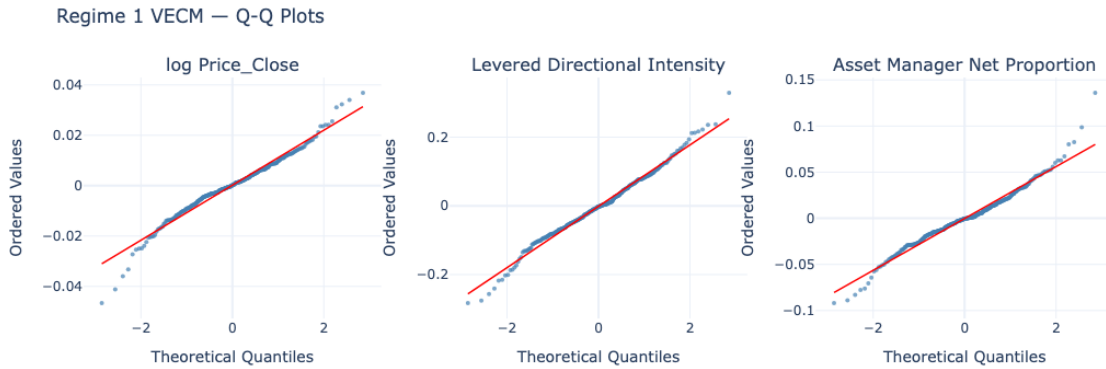


Figure 2: Regime 1 VECM Q-Q plots for Log_Price, Levered Directional Intensity, and Asset Manager Net Proportion residuals. Residuals track the normal line closely through central quantiles; deviations are confined to the tails, indicating fat-tailed rather than misspecified residuals.

4.3.7 Summary

During the YCC stable period, the USD/JPY exchange rate was anchored exclusively in the long run by institutional investor positioning — leveraged fund and asset manager net positions — with macroeconomic fundamentals playing no role. When deviations from equilibrium occurred, positioning adjusted at roughly seven times the speed of the price, consistent with the BoJ absorbing carry trade flows through unlimited bond purchases, leaving traders to reposition around a largely fixed price.

In the short run, changes in the Carry to Risk ratio directly drove both the exchange rate and asset manager positioning, while lagged price movements drove leveraged fund positioning — consistent with trend-following behaviour documented in the CTA literature.

Residual diagnostics confirm correct model specification across all tests.

4.4 Regime 2: Post-Divergence (April 2022 — December 2025)

4.4.1 Variable Selection and Cointegration

Pairwise Engle-Granger cointegration tests within Regime 2 identify JPY Term Spread ($p=0.030$) and log VIX ($p=0.039$) as sharing a long-run relationship with Log_Price. When included together as endogenous variables, log VIX dominates and absorbs the cointegrating relationship, rendering JPY Term Spread insignificant — consistent with the two variables being correlated in the post-divergence period as risk sentiment spikes and BoJ normalisation expectations moved together. JPY Term Spread is retained as the endogenous variable on theoretical grounds, directly capturing market expectations of BoJ policy tightening, while log VIX is moved to the exogenous set as a short-run risk proxy.

The final specification is:

- **Endogenous:** Log_Price, JPY Term Spread
- **Exogenous:** Rate Differential 10Y Delta, Asset Manager Net Proportion Delta, log Nikkei Delta, log VIX Delta

All exogenous variables are in their stationary $I(0)$ form. The rate differential captures the US-Japan yield gap driving the carry trade unwind, asset manager positioning captures institutional flows, log Nikkei Delta captures risk-on equity flows reducing safe-haven yen demand, and log VIX Delta captures short-run safe-haven demand shocks.

4.4.2 Lag Selection and Cointegration Rank

All lag selection criteria unanimously select 0 lags — BIC, AIC, FPE and HQIC all minimise at lag 0. As the VECM requires a minimum of one lag, $k_{ar_diff}=1$ is used.

The Johansen trace test confirms one cointegrating vector at the 90% significance level, with the null of no cointegration ($r \leq 0$) rejected and the null of at most one vector ($r \leq 1$) not rejected. The VECM is estimated with $coint_rank=1$ and $k_{ar_diff}=1$.

Hypothesis	Trace Statistic	CV 90%	CV 95%	CV 99%	Significance
$r \leq 0$	15.247	13.429	15.494	19.935	*
$r \leq 1$	2.582	2.706	3.842	6.635	

* *significant at 10%*

The borderline significance at 90% rather than 95% is noted as a limitation, attributable in part to the smaller sample size of Regime 2 (192 observations) relative to Regime 1 (325 observations). Cointegration tests are known to have reduced power in small samples, and the result is consistent with the pairwise Engle-Granger finding of $p=0.030$ which confirms the long-run relationship independently.

4.4.3 Cointegrating Vector

Variable	Coefficient	Std Error	z-stat	p-value	95% CI
Log Price (β_1)	1.000	—	—	—	—
JPY Term Spread (β_2)	-0.185	0.063	-2.946	0.003***	[-0.308, -0.062]
Constant	-4.884	0.039	-124.409	0.000***	[-4.961, -4.807]

*** significant at 1%

The estimated long-run relationship is:

$$\text{Log_Price} = 4.884 + 0.185 \times \text{JPY Term Spread}$$

The JPY Term Spread is significant at the 1% level ($\beta_2 = -0.185$, $p = 0.003$). The positive coefficient indicates that a steeper Japanese yield curve is associated with a higher USD/JPY — that is, a weaker yen — in the long run. This is consistent with the post-divergence period where a steepening yield curve reflected rising Japanese long-term yields driven by BoJ normalisation, which simultaneously reduced the attractiveness of yen carry trades and signalled a tightening cycle, putting upward pressure on USD/JPY. The result confirms that Japan's yield curve dynamics became the dominant long-run anchor of the exchange rate once YCC was abandoned — a relationship entirely absent in Regime 1.

4.4.4 Error Correction Dynamics

Equation	α (ec1)	Std Error	z-stat	p-value	95% CI
Log Price	-0.060	0.020	-2.986	0.003***	[-0.100, -0.021]
JPY Term Spread	+0.039	0.097	0.405	0.686	[-0.151, 0.230]

*** significant at 1%

The exchange rate error corrects significantly at approximately 6% per week ($\alpha = -0.060$, $p = 0.003$), meaning deviations from the long-run equilibrium are gradually eliminated over time. The JPY Term Spread does not error correct ($p = 0.686$), consistent with it being driven by exogenous BoJ policy decisions rather than adjusting in response to exchange rate dynamics.

The contrast with Regime 1 is the central finding of this analysis. In Regime 1 the exchange rate did not error correct at all — it was the positioning variables that adjusted around a largely fixed price. In Regime 2 the exchange rate itself restores equilibrium, confirming that the structural break at April 2022 marks the transition from a policy-suppressed, sticky exchange rate to a market-determined price that responds to fundamental long-run forces.

4.4.5 Short-Run Dynamics

Panel A: Log_Price equation

Variable	Coefficient	Std Error	z-stat	p-value
Rate Differential 10Y Delta (exog1)	0.020	0.008	2.462	0.014**
Asset Manager Net Proportion Delta (exog2)	-0.179	0.024	-7.483	0.000***
log Nikkei Delta (exog3)	0.097	0.042	2.319	0.020**
log VIX Delta (exog4)	0.009	0.007	1.238	0.216
All lags	—	—	—	n.s.

** significant at 5%, *** significant at 1%

Nothing is significant in the JPY Term Spread equation — the yield curve is driven entirely by exogenous BoJ policy decisions, confirming its role as a long-run anchor rather than a short-run variable.

Three short-run drivers of the exchange rate are significant. Asset Manager Net Proportion Delta is the dominant force (coef=-0.179, p=0.000) — institutional positioning flows have the largest and most significant short-run impact on the yen, consistent across both regimes. The 10-year rate differential (p=0.014) confirms that US-Japan yield dynamics directly drive short-run yen moves in the post-divergence period — a relationship absent in Regime 1 where macro fundamentals were suppressed. Finally, log Nikkei Delta (p=0.020) is positively associated with USD/JPY — rising Japanese equities weaken the yen, consistent with risk-on flows reducing safe-haven demand. VIX is not significant in the short run, suggesting risk sentiment operates through the long-run cointegrating relationship rather than directly.

4.4.6 Diagnostics

Panel A: Univariate Residual Diagnostics

Variable	LB lag 1	LB lag 4	LB lag 8	LB lag 12	JB Stat	JB p-value	Skewness	Kurtosis
Log_Price	0.321	0.639	0.858	0.951	4.22	0.121	-0.336	0.337
JPY Term Spread	0.821	0.878	0.306	0.196	1115.77	0.000***	-0.767	12.122

Panel B: Multivariate Portmanteau Test

Test	Test Statistic	Critical Value (5%)	p-value	Result
Whiteness test (lag 12)	37.635	60.481	0.740	Fail to reject H_0

LB = Ljung-Box p-values. JB = Jarque-Bera. *** significant at 1%

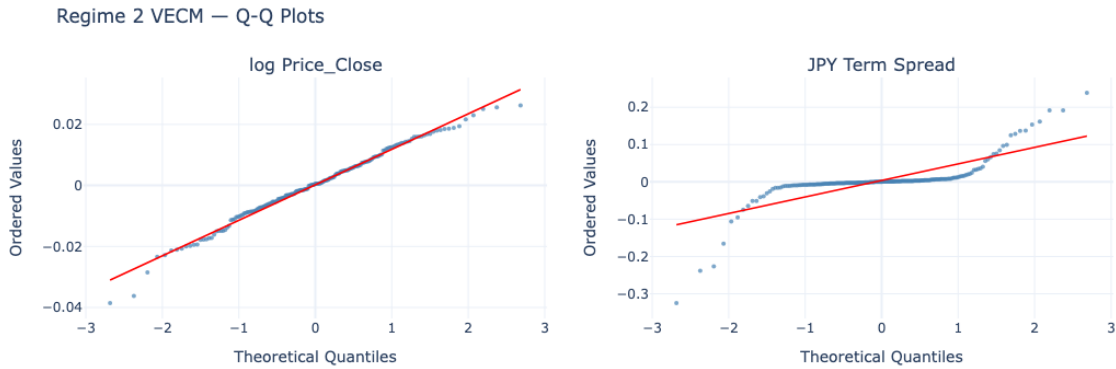


Figure 3: Regime 2 VECM Q-Q plots for Log_Price and JPY Term Spread residuals. Log_Price residuals are near-normal throughout; JPY Term Spread exhibits an S-curve pattern reflecting discrete yield jumps during BoJ policy announcements rather than model misspecification.

All autocorrelation tests pass comfortably. Log_Price residuals are unusually well-behaved — Jarque-Bera fails to reject normality ($p=0.121$), with the Q-Q plot confirming near-normal behaviour across the full distribution. JPY Term Spread exhibits extreme kurtosis (12.12) driven by discrete BoJ policy announcement jumps, visible in the Q-Q plot as a flat central plateau and sharp tail deviations — characteristic of a policy-constrained series rather than model misspecification, confirmed by the passing autocorrelation tests.

4.4.7 Summary

In the post-divergence regime, the USD/JPY exchange rate is anchored in the long run by the JPY term spread — a steeper Japanese yield curve associated with a weaker yen, consistent with BoJ normalisation expectations driving long-end yields higher while simultaneously reducing carry trade attractiveness. Critically, unlike Regime 1, the exchange rate now error corrects significantly at 6% per week, confirming that the structural break restored market-determined exchange rate dynamics.

In the short run, three forces drive the exchange rate: asset manager positioning (dominant, $p=0.000$), the 10-year rate differential ($p=0.014$), and Japanese equity returns ($p=0.020$). The significance of macro fundamentals — entirely absent in Regime 1 — confirms that the post-divergence period represents a genuine regime change in how the yen is priced. Residual diagnostics confirm correct model specification, with Log_Price residuals unusually close to normality and all autocorrelation tests passing comfortably.

4.5. Discussion

4.5.1 Comparing the Two Regimes

The results reveal a fundamental transformation in USD/JPY exchange rate dynamics across the two regimes, summarised below:

Feature	Regime 1 (YCC Stable)	Regime 2 (Post-Divergence)
Long-run anchor	Institutional positioning	JPY term spread
Exchange rate error corrects?	No	Yes (6% per week)
Macro fundamentals significant?	No	Yes
Short-run price driver	Carry to Risk Delta	Asset manager positioning, rate differential, Nikkei
Who adjusts to equilibrium?	Positioning variables	Exchange rate

The contrast is stark. In Regime 1, the BoJ's YCC framework severed the normal link between macroeconomic fundamentals and the exchange rate — the yen was effectively a policy instrument rather than a market price. Institutional investors adapted by anchoring their positioning to the carry trade, with leveraged funds structurally short yen roughly 75% of the time and the Carry to Risk ratio driving both the price and positioning flows. When deviations from equilibrium occurred, it was traders who repositioned rather than the price that adjusted — a direct consequence of the BoJ absorbing carry trade flows through unlimited bond purchases.

In Regime 2, this mechanism breaks down entirely. The long-run anchor shifts to the JPY term spread, reflecting the market's expectation of BoJ policy normalisation. The exchange rate itself now error corrects at 6% per week, macro fundamentals transmit directly to price, and the Nikkei — entirely irrelevant in Regime 1 — becomes a significant short-run driver. The yen transitions from a carry trade vehicle to a conventional floating currency.

4.5.2 Relation to the Broader Literature

UIP and its failure. The finding that interest rate differentials play no role in the long-run level of the exchange rate during Regime 1 is consistent with the extensive literature documenting the failure of Uncovered Interest Parity (UIP). Fama (1984) demonstrates that high interest rate currencies tend to appreciate rather than depreciate as UIP predicts, with the forward premium puzzle reflecting a time-varying risk premium. In the YCC context, this failure is amplified — the BoJ's explicit suppression of Japanese yields rendered the rate differential a poor signal of future exchange rate movements, as any widening differential

simply reflected the policy divergence rather than a genuine market price. Engel (2014) notes that UIP deviations are largest when monetary policy divergence is most extreme, precisely the condition that defined Regime 1. The restoration of UIP-consistent dynamics in Regime 2 — where the rate differential becomes a significant short-run driver — supports Engel's contention that UIP failures are regime-dependent rather than structural.

The carry trade. The Regime 1 findings directly support the carry trade literature. Brunnermeier, Nagel and Pedersen (2009) document that carry trade returns are characterised by periods of gradual appreciation followed by sudden crashes — the "going up by the stairs, coming down by the elevator" phenomenon. The structural short yen positioning of leveraged funds throughout Regime 1, with Levered Directional Intensity averaging -0.221 and net short positions in 75% of weeks, is precisely the crowded carry trade Brunnermeier et al. warn creates systemic crash risk. Galati, Heath and McGuire (2007) provide BIS evidence that yen carry trades were the dominant speculative currency strategy in the pre-crisis period — our results extend this finding through the YCC era, showing that the carry trade remained the structural default position for leveraged money throughout 2016–2022. The April 2022 break represents the beginning of the carry trade unwind they theorise, driven by the Fed hiking cycle making the sustainability of the short yen position increasingly uncertain. Lustig and Verdelhan (2007) link carry trade returns to consumption growth risk, suggesting that the yen's safe-haven appreciation during the August 2024 carry unwind — captured by the post-divergence VIX-yen relationship — reflects exactly the risk rebalancing their framework predicts.

Meese and Rogoff (1983). The seminal finding that structural exchange rate models cannot outperform a random walk out of sample is implicitly addressed by this study's regime-dependent framework. The random walk result of Meese and Rogoff is most likely to hold when exchange rate dynamics are structurally unstable — precisely what the Bai-Perron structural break confirms. Within each regime the VECM finds stable, significant long-run relationships, suggesting that the failure to beat a random walk in full-sample models reflects regime mixing rather than a fundamental absence of exchange rate predictability. MacDonald and Taylor (1994) make a similar argument for the monetary model, finding that long-run relationships are recoverable once non-stationarity and cointegration are properly accounted for — our results support this view within the regime-specific framework.

4.5.3 Methodological Contribution

The use of Johansen (1988) cointegration within a regime-specific VECM framework proves essential to uncovering these dynamics. A full-sample VECM — as estimated in the earlier literature — would conflate the two regimes and produce a misleading picture of exchange rate determination. Gregory and Hansen (1996) extend cointegration testing to allow for regime shifts, arguing that standard cointegration tests have low power in the presence of structural breaks. This study addresses this directly by endogenously identifying breaks via Bai-Perron before conducting regime-specific cointegration tests, yielding cleaner and more economically interpretable results than a single full-sample model would produce. The GARCH(1,1) specification following Bollerslev (1986) and Baillie and Bollerslev (1989) captures the volatility persistence documented in both regimes — particularly the contrast between the suppressed but persistent volatility of Regime 1 ($\beta=0.797$) and the shock-driven volatility of the post-divergence period.

4.5.4 Overall Conclusion

This study investigates the determinants of the USD/JPY exchange rate across two structurally distinct regimes identified by an endogenous Bai-Perron structural break at April 2022. The results deliver a coherent and empirically robust narrative: the Bank of Japan's Yield Curve Control policy fundamentally suppressed the normal transmission mechanism between macroeconomic fundamentals and the exchange rate, leaving institutional carry trade positioning as the sole long-run anchor of the yen. The Federal Reserve's rate hiking cycle in April 2022 — by threatening the long-run sustainability of the carry trade — triggered a regime change in exchange rate dynamics, confirmed by the Chow test F-statistic of 122.76. In the post-divergence regime, the yen transitions to being anchored by Japan's yield curve, with the exchange rate now responding to rate differentials, equity flows, and institutional positioning in the manner predicted by standard open economy models.

These findings contribute to three strands of the literature. They provide regime-specific evidence on UIP failure, showing that deviations are policy-induced rather than structural. They document the persistence and eventual unwind of the yen carry trade, extending the findings of Brunnermeier et al. (2009) through the YCC era. And they demonstrate that exchange rate predictability — elusive in full-sample models since Meese and Rogoff (1983) — is recoverable within stable regimes when structural breaks are properly identified. The broader implication is that central bank yield curve intervention does not merely affect the level of interest rates — it fundamentally alters the architecture of exchange rate determination, with consequences that persist well beyond the formal policy period.

Appendix A: Unit Root Test Results

Unit root tests are conducted using the Augmented Dickey-Fuller (ADF) and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests. A variable is classified as stationary $I(0)$ only if the ADF rejects the unit root null ($p < 0.05$) and KPSS fails to reject the stationarity null ($p > 0.05$). All level variables are confirmed $I(1)$ and all transformed variables are confirmed $I(0)$.

Table A1: ADF and KPSS Unit Root Test Results

Variable	Form	ADF Stat	ADF p	KPSS Stat	KPSS p	Order
Price Close	Level	-0.203	0.938	2.598	0.010	I(1)
JPY Term Spread	Level	0.006	0.959	2.866	0.010	I(1)
Levered Directional Intensity	Level	-3.789	0.003	0.698	0.014	I(1)
Asset Manager Net Proportion	Level	-2.795	0.059	0.184	0.100	I(1)
Rate Differential 3M	Level	-1.818	0.371	1.670	0.010	I(1)
Rate Differential 10Y	Level	-1.664	0.450	0.900	0.010	I(1)
Carry to Risk	Level	-1.526	0.521	1.270	0.010	I(1)
log Oil	Level	-2.547	0.105	1.381	0.010	I(1)
log VIX	Level	-3.750	0.004	0.770	0.010	I(1)
log Nikkei	Level	-0.238	0.934	3.048	0.010	I(1)
log Price Close Delta	Transformed	-22.290	0.000	0.229	0.100	I(0)
JPY Term Spread Delta	Transformed	-9.403	0.000	0.108	0.100	I(0)
Levered Directional Intensity Delta	Transformed	-7.818	0.000	0.017	0.100	I(0)
Asset Manager Net Proportion Delta	Transformed	-18.747	0.000	0.045	0.100	I(0)
Rate Differential 3M Delta	Transformed	-3.126	0.025	0.233	0.100	I(0)
Rate Differential 10Y Delta	Transformed	-8.825	0.000	0.120	0.100	I(0)
Carry to Risk Delta	Transformed	-6.288	0.000	0.104	0.100	I(0)
log Oil Delta	Transformed	-8.842	0.000	0.081	0.100	I(0)
log VIX Delta	Transformed	-12.863	0.000	0.027	0.100	I(0)
log Nikkei Delta	Transformed	-23.718	0.000	0.105	0.100	I(0)

ADF null: unit root present. KPSS null: series is stationary. I(1) classification requires ADF $p > 0.05$ or KPSS $p < 0.05$. I(0) classification requires ADF $p < 0.05$ and KPSS $p > 0.05$. KPSS p-values reported as 0.010 indicate the test statistic exceeds the upper bound of the look-up table; the actual p-value is at or below 0.010.

Appendix B: Pairwise Engle-Granger Cointegration Tests

Pairwise Engle-Granger cointegration tests are conducted between log USD/JPY and each candidate I(1) variable. Tests are run on the full sample and separately within each regime. The null hypothesis is no cointegration.

Table B1: Full Sample (Jan 2016 — Dec 2025, N=517)

Variable	EG Statistic	p-value	Significance
JPY Term Spread	-3.796	0.014	**
Rate Differential 3M	-1.473	0.771	
Rate Differential 10Y	-0.418	0.970	
Carry to Risk	-1.095	0.883	
Asset Manager Net Proportion	0.042	0.987	
Levered Directional Intensity	-1.025	0.898	
log Oil	-1.399	0.798	
log VIX	-0.438	0.969	
log Nikkei	-2.253	0.397	

Table B2: Regime 1 — YCC Stable (Jan 2016 — Apr 2022, N=326)

Variable	EG Statistic	p-value	Significance
JPY Term Spread	-3.047	0.099	*
Rate Differential 3M	-2.618	0.230	
Rate Differential 10Y	-2.957	0.121	
Carry to Risk	-2.581	0.245	
Asset Manager Net Proportion	-3.236	0.064	*
Levered Directional Intensity	-3.788	0.014	**
log Oil	-3.448	0.037	**
log VIX	-2.586	0.243	
log Nikkei	-2.745	0.184	

Table B3: Regime 2 — Post-Divergence (Apr 2022 — Dec 2025, N=192)

Variable	EG Statistic	p-value	Significance
JPY Term Spread	-3.623	0.023	**
Rate Differential 3M	-1.950	0.554	
Rate Differential 10Y	-2.170	0.440	
Carry to Risk	-2.416	0.317	
Asset Manager Net Proportion	-2.680	0.207	
Levered Directional Intensity	-2.724	0.191	
log Oil	-2.802	0.165	
log VIX	-3.099	0.089	*
log Nikkei	-3.750	0.016	**

** significant at 10%, ** significant at 5%, *** significant at 1%. EG = Engle-Granger test statistic. Null hypothesis: no cointegration.*

The full-sample results identify only JPY Term Spread as cointegrated with log USD/JPY ($p=0.014$). The regime-specific results reveal that the cointegrating variables change across regimes — in Regime 1, Levered Directional Intensity and log Oil are cointegrated with the exchange rate, while in Regime 2 these relationships disappear and JPY Term Spread and log Nikkei emerge as the significant long-run anchors. This regime-dependence of cointegrating relationships motivates the separate VECM estimation and is a central finding of the study.

Appendix C: Full Short-Run Dynamics Tables

C1: Regime 1 — YCC Stable (Jan 2016 — Apr 2022)

Panel A: Equation — Log_Price

Variable	Coefficient	Std Error	z-stat	p-value
Carry to Risk Delta (exog1)	0.175	0.049	3.597	0.000***
log Oil Delta (exog2)	-0.005	0.010	-0.508	0.611
L1.Log_Price	-0.028	0.062	-0.446	0.656
L1.Levered Directional Intensity	-0.004	0.007	-0.571	0.568
L1.Asset Manager Net Proportion	-0.011	0.024	-0.457	0.648
L2.Log_Price	-0.064	0.063	-1.020	0.308
L2.Levered Directional Intensity	0.015	0.006	2.290	0.022**
L2.Asset Manager Net Proportion	-0.005	0.024	-0.195	0.845

Panel B: Equation — Levered Directional Intensity

Variable	Coefficient	Std Error	z-stat	p-value
Carry to Risk Delta (exog1)	-0.020	0.394	-0.050	0.960
log Oil Delta (exog2)	0.006	0.081	0.072	0.943
L1.Log_Price	-1.992	0.506	-3.938	0.000***
L1.Levered Directional Intensity	0.055	0.057	0.964	0.335
L1.Asset Manager Net Proportion	0.460	0.194	2.373	0.018**
L2.Log_Price	-1.344	0.509	-2.643	0.008***
L2.Levered Directional Intensity	-0.078	0.052	-1.499	0.134
L2.Asset Manager Net Proportion	0.013	0.194	0.067	0.947

Panel C: Equation — Asset Manager Net Proportion

Variable	Coefficient	Std Error	z-stat	p-value
Carry to Risk Delta (exog1)	-0.350	0.125	-2.797	0.005***
log Oil Delta (exog2)	0.007	0.026	0.262	0.794

L1.Log_Price	-0.071	0.161	-0.440	0.660
L1.Levered Directional Intensity	0.010	0.018	0.540	0.589
L1.Asset Manager Net Proportion	0.135	0.062	2.196	0.028**
L2.Log_Price	0.079	0.162	0.489	0.625
L2.Levered Directional Intensity	0.017	0.017	1.002	0.316
L2.Asset Manager Net Proportion	0.024	0.062	0.383	0.702

C2: Regime 2 — Post-Divergence (Apr 2022 — Dec 2025)

Panel A: Equation — Log_Price

Variable	Coefficient	Std Error	z-stat	p-value
Rate Differential 10Y Delta (exog1)	0.020	0.008	2.462	0.014**
Asset Manager Net Proportion Delta (exog2)	-0.179	0.024	-7.483	0.000***
log Nikkei Delta (exog3)	0.097	0.042	2.319	0.020**
log VIX Delta (exog4)	0.009	0.007	1.238	0.216
L1.Log_Price	-0.076	0.063	-1.214	0.225
L1.JPY Term Spread	0.016	0.015	1.044	0.297

Panel B: Equation — JPY Term Spread

Variable	Coefficient	Std Error	z-stat	p-value
Rate Differential 10Y Delta (exog1)	-0.062	0.040	-1.567	0.117
Asset Manager Net Proportion Delta (exog2)	-0.069	0.116	-0.592	0.554
log Nikkei Delta (exog3)	0.197	0.201	0.982	0.326
log VIX Delta (exog4)	0.010	0.033	0.292	0.771
L1.Log_Price	0.138	0.303	0.455	0.649
L1.JPY Term Spread	-0.001	0.074	-0.013	0.990

** significant at 5%, *** significant at 1%

Appendix D: GARCH(1,1) Volatility Model Results

A GARCH(1,1) model with Student's t distributed errors is fitted to the Log_Price residuals from each regime VECM to characterise volatility dynamics. The specification follows Bollerslev (1986): $\sigma_t^2 = \omega + \alpha \cdot \varepsilon_{t-1}^2 + \beta \cdot \sigma_{t-1}^2$.

Table D1: GARCH(1,1) — Regime 1 (YCC Stable, Jan 2016 — Apr 2022) $N = 322$ observations

Parameter	Coefficient	Std Error	t-stat	p-value	Sig
ω (omega)	0.091	0.106	0.854	0.393	
α (alpha)	0.132	0.110	1.203	0.229	
β (beta)	0.797	0.175	4.546	0.000	***
ν (t-dist)	5.664	1.784	3.176	0.002	***

Volatility persistence ($\alpha + \beta$) = 0.928 | Half-life of volatility shock = 9.3 weeks

Table D2: GARCH(1,1) — Regime 2 (Post-Divergence, Apr 2022 — Dec 2025) $N = 190$ observations

Parameter	Coefficient	Std Error	t-stat	p-value	Sig
ω (omega)	0.044	0.037	1.200	0.230	
α (alpha)	0.078	0.027	2.911	0.004	***
β (beta)	0.888	0.026	34.066	0.000	***
ν (t-dist)	14.586	12.317	1.184	0.236	

Volatility persistence ($\alpha + \beta$) = 0.966 | Half-life of volatility shock = 19.9 weeks***
significant at 1%. ω = baseline variance; α = ARCH effect (sensitivity to recent shocks); β = GARCH effect (volatility persistence); ν = degrees of freedom of Student's t distribution.

Interpretation: Both regimes exhibit high volatility persistence. In Regime 1, the ARCH effect is insignificant while β is strongly significant (0.797), indicating volatility was persistent but not shock-driven — consistent with BoJ suppression of volatility spikes under YCC. In Regime 2, both α (0.078, $p=0.004$) and β (0.888, $p=0.000$) are significant, indicating volatility now responds to shocks and persists — consistent with the more turbulent post-divergence environment. Persistence increases from 0.928 to 0.966 across regimes, with the half-life of a volatility shock rising from 9.3 to 19.9 weeks.

Appendix E: Residual Diagnostic Plots

E1: Regime 1 — YCC Stable (Jan 2016 — Apr 2022)

Figure E1: Regime 1 VECM — Residuals Over Time and Distribution

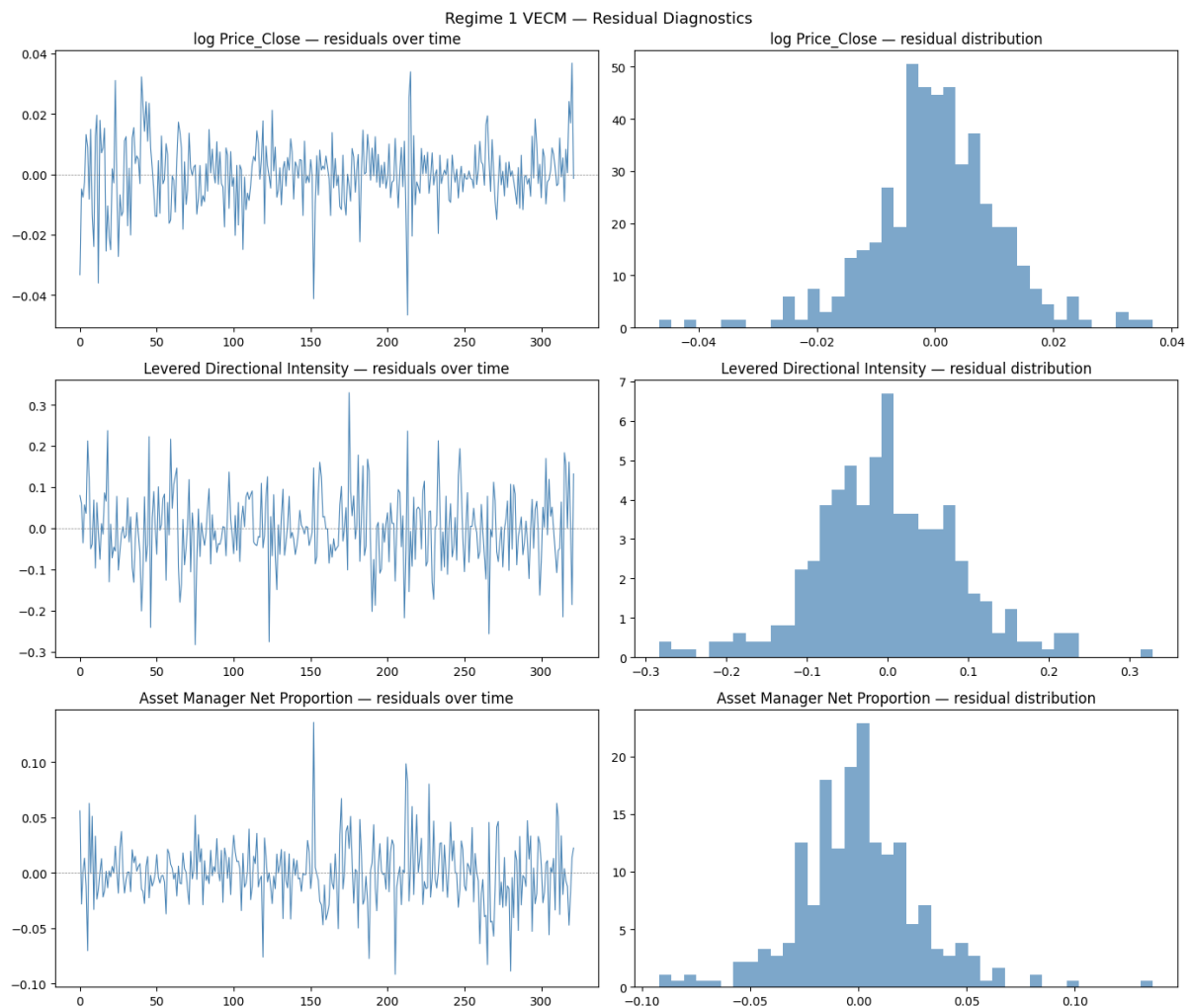


Figure E1: Regime 1 VECM residuals over time (left) and residual distributions (right) for Log_Price, Levered Directional Intensity, and Asset Manager Net Proportion. Residuals are centred around zero with no systematic patterns. The spike around observation 210 in Log_Price corresponds to the COVID-19 shock (March 2020).

Figure E2: Regime 1 VECM — Q-Q Plots

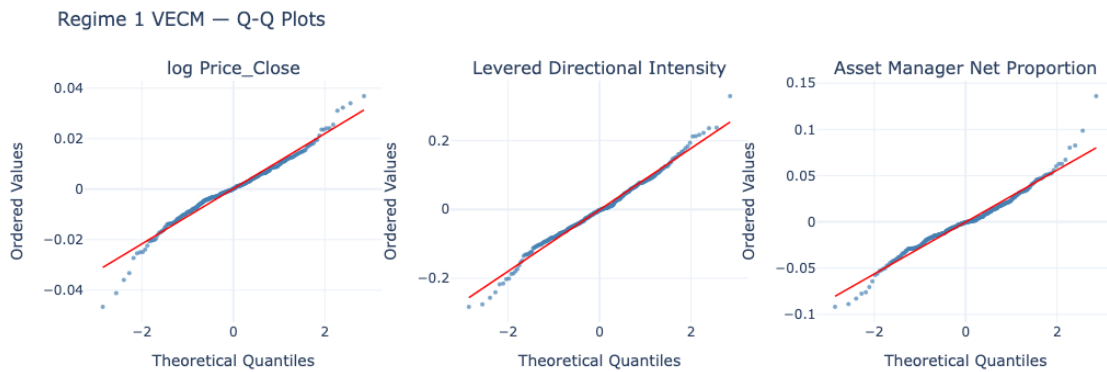


Figure E2: Regime 1 VECM Q-Q plots for Log_Price, Levered Directional Intensity, and Asset Manager Net Proportion residuals. All three series track the normal line closely through the central quantiles, with fat-tail deviations only at the extremes.

E2: Regime 2 — Post-Divergence (Apr 2022 — Dec 2025)

Figure E3: Regime 2 VECM — Residuals Over Time and Distribution

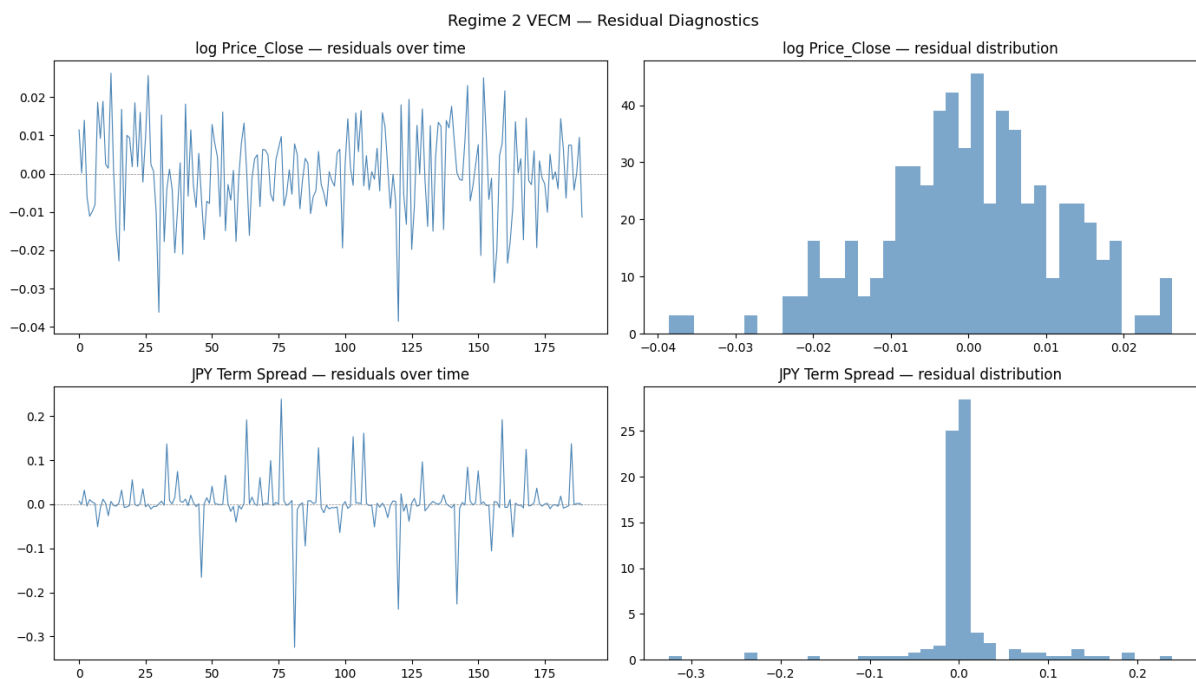


Figure E3: Regime 2 VECM residuals over time (left) and residual distributions (right) for Log_Price and JPY Term Spread. Log_Price residuals are well-behaved; JPY Term Spread shows isolated spikes corresponding to BoJ policy announcement dates rather than systematic misspecification.

Figure E4: Regime 2 VECM — Q-Q Plots

